

Prognostic impact of respiratory sarcopenia in older patients with heart failure: a *post hoc* analysis from the SONIC-HF registry

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Aims

Respiratory sarcopenia, characterized by reduced respiratory muscle mass and function, can impair ventilatory reserve and physical capacity, potentially worsening outcomes. However, its prevalence and prognostic significance in older patients with heart failure remain unclear. This study aimed to investigate the clinical significance of respiratory sarcopenia in this population.

Methods and results

This was a *post hoc* analysis of the compariSON of various methods In evaluation of sarCopenia in patients with Heart Failure (SONIC-HF) study, a multicentre prospective observational study. Among 435 patients hospitalized for heart failure [median age: 81 (interquartile range 74–85) years; 41.8% female], we defined respiratory sarcopenia as the presence of both low resting diaphragm thickness, assessed using ultrasonography, and reduced per cent predicted forced vital capacity (FVC), a surrogate for respiratory muscle strength. The primary outcome was 2 year all-cause mortality. Respiratory sarcopenia was observed in 47 patients (10.8%). During the 2 year follow-up, 78 patients (17.9%) died. All-cause mortality was significantly higher among patients with respiratory sarcopenia than those without ($P < 0.001$). In Cox proportional hazards analysis, respiratory sarcopenia was independently associated with increased mortality risk (hazard ratio, 2.51; 95% confidence interval, 1.40–4.50; $P = 0.002$), even after adjustment for conventional risk factors. Low diaphragm thickness or reduced per cent predicted FVC alone were not associated with mortality.

Conclusion

In older patients with heart failure, respiratory sarcopenia was uncommon but associated with significantly higher mortality. Simultaneous assessment of respiratory muscle mass and function may aid in identifying high-risk individuals and enhance risk stratification beyond structural assessments alone.

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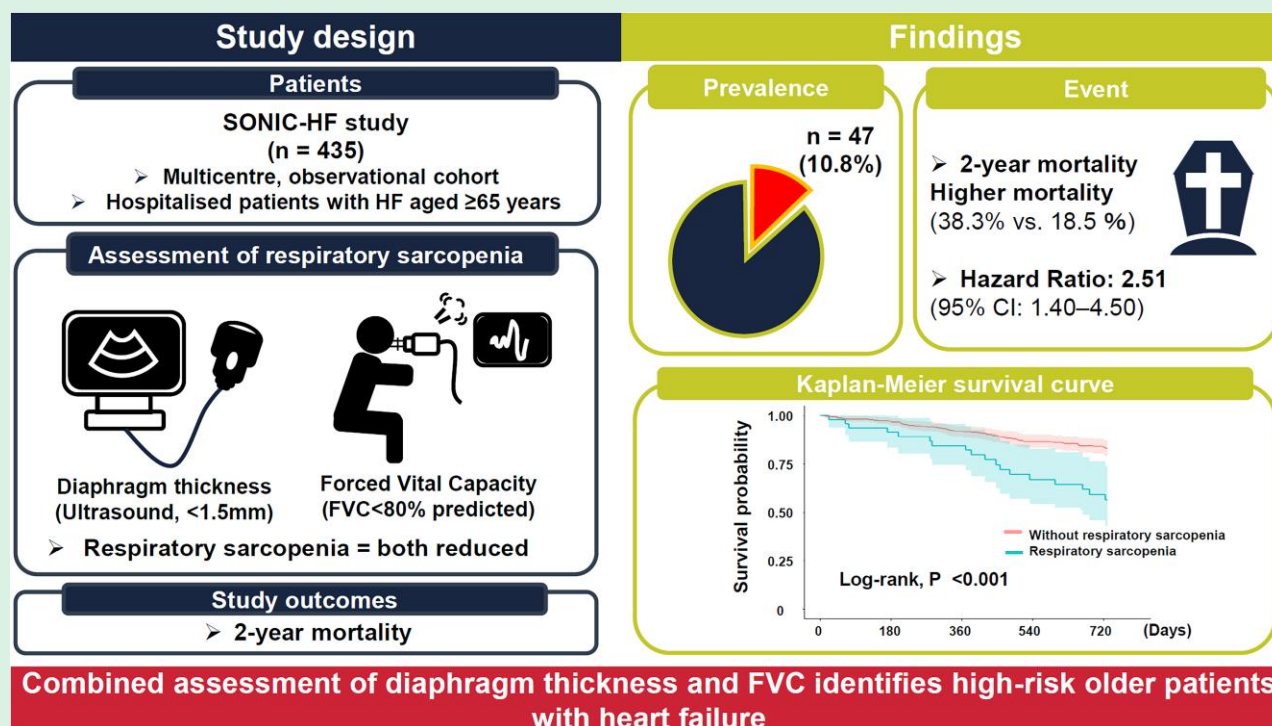
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Lay summary

This study found that ‘respiratory sarcopenia’—a condition in which both respiratory muscle mass and strength are reduced—is a strong predictor of poor survival in older patients hospitalized for heart failure.

- Respiratory sarcopenia was present in approximately 10% of hospitalized heart failure patients aged 65 and older, and it was linked to more than double the risk of death over 2 years.
- The condition can be easily detected using simple bedside tests (ultrasound and spirometry), suggesting that it could be a practical and valuable tool for identifying high-risk patients.

Graphical Abstract



Keywords

Respiratory sarcopenia • Diaphragm thickness • Forced vital capacity • Ultrasonography • Heart failure • Prognosis

Introduction

As the global population continues to age, heart failure has become a growing public health concern, with its incidence and poor prognosis closely tied to advanced age.^{1–3} Sarcopenia, a condition characterized by progressive loss of skeletal muscle mass and strength, is also age-related and has been recognized as a key extracardiac determinant of adverse outcomes in patients with heart failure.⁴ Breathing is a vital function sustained by the respiratory muscles. Impairments in respiratory muscle function may reduce ventilatory reserve, contributing to exertional dyspnoea, exercise intolerance, and decreased physical activity, ultimately worsening clinical prognosis.⁵ Respiratory sarcopenia refers to a condition characterized by reductions in both respiratory muscle mass and strength. A recent joint statement by the Japanese Society for Respiratory Care and Rehabilitation, the Japanese Association on Sarcopenia and Frailty, the Japanese Society of Respiratory Physical Therapy,

and the Japanese Association of Rehabilitation Nutrition formally introduced the concept of respiratory sarcopenia and proposed standardized diagnostic criteria.⁵ Consequently, respiratory sarcopenia is increasingly recognized as a clinically significant condition, further underscoring its relevance in clinical practice. Importantly, unlike the diagnosis of systemic sarcopenia, respiratory sarcopenia can be assessed by measuring diaphragm thickness as a surrogate for respiratory muscle mass, which can be easily and non-invasively evaluated using widely available ultrasonography.^{6,7} However, owing in part to the absence of routine diaphragm assessment in daily clinical practice, little information exists regarding the prevalence and prognostic significance of respiratory sarcopenia in older patients with heart failure, despite the availability of diagnostic criteria. Therefore, this study aimed to investigate the prevalence of respiratory sarcopenia and its impact on long-term outcomes in older patients hospitalized for heart failure by conducting a *post hoc* analysis of the compariSON of various methods in

evaluation of sarcopenia in patients with Heart Failure (SONIC-HF) cohort—a prospective, multicentre observational registry that includes ultrasound-based assessment of skeletal muscles in this patient population.

Methods

Study design and patient population

This study represented a *post hoc* analysis of the SONIC-HF study, a prospective, multicentre observational cohort study conducted to evaluate the prognostic value of muscle mass and function using ultrasonography and anthropometric measurements.^{8,9} Consecutive patients hospitalized for decompensated heart failure, aged ≥ 65 years and ambulatory at the time of assessment, were enrolled between March 2018 and December 2019. The diagnosis of heart failure decompensation was based on the Framingham criteria. Patients were excluded if any of the following applied: (i) previous heart transplantation or use of a left ventricular assist device; (ii) current treatment with long-term haemodialysis or peritoneal dialysis; (iii) diagnosis of acute myocarditis; (iv) missing data for B-type natriuretic peptide (BNP) or N-terminal pro-BNP (NT-proBNP); or (v) BNP <100 pg/mL or NT-proBNP <300 pg/mL at the time of admission. Among the four participating centres in Japan, one was a university hospital and three were non-university teaching hospitals. Clinical, echocardiographic, and laboratory data were collected while patients were in a stable condition before hospital discharge. Prediabetes was defined as a glycated haemoglobin level of 5.7–6.4% in patients without diabetes.¹⁰ The study was conducted in accordance with the Declaration of Helsinki and was approved by the institutional review boards of all participating centres. In accordance with the Japanese Ethical Guidelines for Medical and Health Research Involving Human Subjects, written informed consent was not required due to the observational nature of the study and the absence of invasive procedures. Participants were informed of their right to withdraw at any time. Study details, including objectives, inclusion and exclusion criteria, and participating centres, were publicly disclosed on the University Hospital Medical Information Network (UMIN000031635) before patient enrolment.

Assessment of respiratory sarcopenia

Respiratory sarcopenia was defined based on decreased respiratory muscle quantity and strength, in accordance with previously published literature.⁵ Diaphragm muscle quantity was assessed using resting diaphragm thickness measured by ultrasonography, whereas respiratory muscle strength was evaluated using forced vital capacity (FVC), which correlates well with sniff nasal inspiratory pressure.^{11,12} Resting diaphragm thickness was measured by trained and certified observers using ultrasonography. Measurements were obtained at the lowest intercostal space between the right anterior and mid-axillary lines, with patients in a stable condition following completion of intravenous therapy. A high-frequency linear probe (7.5–13 MHz) was used to measure diaphragm thickness twice during natural expiration, and the mean value was recorded. To ensure consistency and reliability, all observers completed a structured ultrasound certification programme. Initial practice was performed on volunteers, and image submissions were reviewed by a central core laboratory for quality assessment. Certification to assess study participants was granted only to those who demonstrated reproducible and accurate measurements. Interobserver variability was previously validated in the SONIC-HF study using 90 randomly selected images, showing acceptable agreement between blinded observers.⁹ Pulmonary function was measured in the seated position using an electronic spirometer, following standard pulmonary function test protocols.¹¹ For FVC measurement,

patients were instructed to take a full inspiration followed by a maximally forced expiration, ensuring an airtight seal and sustained effort until the end of exhalation. The highest value from at least three technically acceptable trials was recorded. The percentage of the predicted FVC (%FVC) and other parameters was calculated using sex-, age-, and height-specific reference equations provided by the Japanese Respiratory Society.¹³ A resting diaphragm thickness <0.15 cm, based on the lower limit observed in healthy adults,^{5,14} and a %FVC $<80\%$ served as cut-off values.¹⁵ Patients meeting both criteria were diagnosed with respiratory sarcopenia.

Assessment of physical function

Physical function was evaluated through measurements of muscle mass, muscle strength, physical performance, and the 6 min walk test (6MWT).^{6,16} Muscle mass was assessed using bioelectrical impedance analysis (BIA; BC-622, Tanita, Tokyo, Japan), with appendicular skeletal muscle mass calculated as the sum of arm and leg muscle mass. The appendicular skeletal muscle mass index was derived by dividing appendicular skeletal muscle mass by height squared (kg/m^2). Muscle strength was measured using handgrip strength. Participants were seated with their elbow flexed at 90° , and two trials were performed alternately for each hand; the highest value across trials was recorded as the maximal grip strength (kg).⁶ Physical performance was assessed using the Short Physical Performance Battery (SPPB), comprising standing balance, gait speed, and chairstand tests, with scores ranging from 0 to 12.¹⁷ The 6MWT was performed according to standardized protocols by trained physical therapists and/or heart failure specialists after completion of decongestion therapy and clinical stabilization. Participants were instructed to walk as far as possible along a 30 m course for 6 min; assistive devices were permitted if necessary.¹⁸ The cut-off values for these indicators were determined according to the Asian Working Group for Sarcopenia's (AWGS) 2019 Consensus Update and previous research.⁶ The thresholds for the appendicular skeletal muscle mass index were <7.0 kg/m^2 for men and <5.7 kg/m^2 for women.⁶ The cut-off values for muscle strength were <28 kg for men and <18 kg for women.⁶ Physical performance was also evaluated using parameters defined in the AWGS 2019 criteria, which provide validated cut-offs for gait speed, the five-time chair-stand test, and the SPPB.⁶ The cut-off value for the 6 min walk distance (6MWD) was set at <300 m, based on previous literature.^{16,19}

Follow-up and study outcome

The primary outcome of this study was all-cause mortality within 2 years following hospital discharge. Patient prognoses were prospectively tracked for up to 2 years post-discharge, with follow-up concluding in December 2021. Most patients continued routine follow-up at outpatient clinics, from which prognostic data were collected using hospital records. For patients lacking follow-up data in hospital records, survival status was confirmed via telephone interviews with family members. During follow-up, all deaths were classified into eight specific categories according to definitions established in previous studies.^{20,21} These categories included death due to heart failure, acute coronary syndrome, sudden cardiac death, stroke, renal failure, other cardiovascular causes, non-cardiovascular or non-renal causes, and unknown causes. To further evaluate the association between sarcopenia and cause-specific mortality, these causes were further grouped into three broader categories: (i) heart failure-related deaths (including deaths classified as heart failure death), (ii) other cardiovascular deaths (comprising deaths due to acute coronary syndrome, sudden cardiac death, stroke, renal failure, and other cardiovascular causes), and (iii) non-cardiovascular deaths (including deaths due to non-cardiovascular or non-renal causes and

unknown deaths). Detailed clinical circumstances surrounding each death were also collected. Cause-of-death adjudication was independently performed by two board-certified cardiologists who were not involved in the original SONIC-HF study. Any discrepancies were resolved through discussion with a third cardiologist.

Statistical analysis

Categorical variables were summarized as frequencies and percentages and compared using the χ^2 test or Fisher's exact test, as appropriate. Continuous variables were expressed as means with standard deviations if normally distributed, or as medians with interquartile ranges (IQRs) if non-normally distributed. For comparisons between the two groups, Student's *t*-test was used for normally distributed variables, and the Mann–Whitney U test was used for non-normally distributed variables. Multivariable logistic regression models were constructed to evaluate the associations between respiratory sarcopenia and physical function parameters. Covariates included age and sex, in accordance with previous literature.²² Kaplan–Meier survival curves were generated to assess differences in mortality, with significance determined using the log-rank test. Cox proportional hazards models were used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs) for the association between respiratory sarcopenia and 2 year all-cause mortality. In the multivariate Cox model, adjustments were made for the Meta-analysis Global Group in Chronic Heart Failure (MAGGIC) risk score and log-transformed BNP levels. The MAGGIC score is a well-established predictor of mortality in patients with heart failure,²³ and the inclusion of BNP has been shown to enhance prognostic accuracy in Japanese cohorts.²⁴ Sensitivity analysis based on univariate and multivariable Cox regression analyses was additionally performed. Variables with $P < 0.05$ in the univariate analyses, together with respiratory sarcopenia, were subsequently included in the multivariable model.

A two-sided P -value of <0.05 was considered statistically significant. All statistical analyses were conducted using R Studio, version 4.2.2 (R Foundation for Statistical Computing, Vienna, Austria).

Results

Of the 692 inpatients originally enrolled in the SONIC-HF study, 257 were excluded owing to missing data on diaphragm thickness or FVC, leaving 435 patients [median age: 81 (IQR: 74–85) years; 41.8% female] for the final analysis (see [Supplementary material online, Figure S1](#)). Among these, respiratory sarcopenia was identified in 47 patients (10.8%). The baseline characteristics of patients stratified by respiratory sarcopenia status are presented in [Table 1](#). Compared with patients without respiratory sarcopenia, those with respiratory sarcopenia were older, had a lower body mass index, and a higher proportion of New York Heart Association (NYHA) Class III or IV. No significant differences in the prevalence of hypertension or the distribution of diabetes status were observed between groups. Physical function was consistently lower in the respiratory sarcopenia group across all assessed measures, including the SPPB total score, 4 m gait speed, five-time chair-stand test, 6MWD, and handgrip strength, compared with the group without respiratory sarcopenia ([Table 2](#)). The respiratory sarcopenia group also had decreased pulmonary function ([Table 2](#)). Although both low handgrip strength and 6MWD <300 m were significantly associated with respiratory sarcopenia in the univariate analysis, none of the sarcopenia-related markers remained statistically significant in the logistic regression models

after adjustment for age and sex (see [Supplementary material online, Table S1](#)).

During the 2 year follow-up period (follow-up completion rate: 98.4%), 78 patients (17.9%) died: 60 deaths (15.5%) occurred among patients without respiratory sarcopenia and 18 (38.3%) among those with respiratory sarcopenia. When examining cause-specific mortality, patients with respiratory sarcopenia showed no apparent differences in the proportions of heart failure-related deaths, other cardiovascular deaths, or non-cardiovascular deaths from those without respiratory sarcopenia. In both groups, non-cardiovascular deaths accounted for the majority of cases (see [Supplementary material online, Table S2](#)). Kaplan–Meier survival analysis demonstrated significantly lower survival rates in patients with respiratory sarcopenia compared with those without (log-rank test, $P < 0.001$; [Figure 1](#)). To further explore the prognostic relevance across heart failure phenotypes, subgroup analyses were conducted according to the left ventricular ejection fraction (LVEF $<50\%$ and $\geq 50\%$). In both subgroups, patients with respiratory sarcopenia consistently exhibited significantly lower survival rates (log-rank test, $P = 0.02$ and $P = 0.001$, respectively; [Supplementary material online, Figure S2](#)). In the Cox proportional hazard analysis, after adjusting for the MAGGIC score and log-transformed BNP, respiratory sarcopenia remained independently associated with an increased 2-year mortality risk (HR, 2.51; 95% CI, 1.40–4.50; $P = 0.002$; [Table 3](#)). After further adjustment in the model, no significant interaction was observed between LVEF $<50\%$ and $\geq 50\%$ regarding the prognostic impact of respiratory sarcopenia (P for interaction = 0.676). For a sensitivity analysis, we additionally conducted multivariable Cox regression analyses incorporating all baseline variables with $P < 0.05$ in the univariate analyses. Respiratory sarcopenia remained an independent predictor of 2 year all-cause mortality in this model [HR, 2.41 (95% CI, 1.31–4.42); $P = 0.005$; [Supplementary material online, Table S3](#)]. Further sensitivity analyses were conducted by adding chronic obstructive pulmonary disease (COPD), forced expiratory volume in one second (FEV₁)/FVC ratio, and predicted FEV₁ (%FEV₁) to the model, along with the MAGGIC score and log-transformed BNP. Respiratory sarcopenia consistently remained an independent predictor of 2 year all-cause mortality across these models [HR, 2.51 (95% CI, 1.40–4.50); $P = 0.002$; HR, 2.56 (95% CI, 1.43–4.60); $P = 0.002$; and HR, 2.31 (95% CI, 1.17–4.58); $P = 0.016$, respectively].

To examine the respective contributions of respiratory muscle mass and strength, patients without respiratory sarcopenia were subdivided into three groups: those with both diaphragm thickness and FVC preserved (normal group: $n = 187$), those with low diaphragm thickness and preserved FVC (isolated diaphragm atrophy group, $n = 32$), and those with preserved diaphragm thickness and reduced FVC (isolated FVC reduction group, $n = 169$). The respiratory sarcopenia group ($n = 47$) was included as the fourth group. Kaplan–Meier curves (see [Supplementary material online, Figure S3](#)) indicated a significant difference in prognosis among the four groups, and the isolated FVC reduction group showed a numerically lower survival rate. However, in unadjusted and adjusted Cox proportional hazards models, only the respiratory sarcopenia group, but neither the isolated diaphragm atrophy group (HR: 1.16, 95% CI: 0.66–2.01; $P = 0.612$) nor the isolated FVC reduction group (HR,

Table 1 Baseline characteristics of the study population

Variables	Without respiratory sarcopenia (n = 388)	Respiratory sarcopenia (n = 47)	P-value	Missing rate (%)
Age (years)	80 (73–85)	84 (80–87)	<0.001	0.0
Male sex (%)	231 (59.7)	21 (44.7)	0.060	0.1
NYHA class III/IV (%)	37 (9.6)	13 (27.7)	0.001	1.0
BMI (kg/m ²)	22.0 ± 3.8	20.1 ± 3.4	0.001	9.4
SBP (mm Hg)	114 ± 17	110 ± 15	0.082	0.3
DBP (mm Hg)	63 ± 10	61 ± 10	0.207	0.3
Heart rate (bpm)	71 ± 13	74 ± 15	0.134	0.3
LVEF (%)	46 ± 16	51 ± 18	0.046	6.5
History of heart failure (%)			0.117	0.1
De-novo	176 (45.5)	20 (42.6)		
Less than 1.5 years	68 (17.6)	14 (29.8)		
Equalling/more than 1.5 years	143 (37.0)	13 (27.7)		
Comorbidities (%)				
Hypertension	238 (61.5)	29 (61.7)	>0.999	0.1
Diabetes status			0.596	10.6
Normoglycaemia	89 (25.5)	13 (32.5)		
Prediabetes	106 (30.4)	12 (30.0)		
Diabetes	154 (44.1)	15 (37.5)		
Atrial fibrillation	202 (52.2)	24 (51.1)	>0.999	0.1
COPD	38 (9.8)	7 (14.9)	0.308	0.1
Coronary artery disease	144 (37.3)	15 (31.9)	0.524	0.3
Prescription at discharge (%)				
Loop diuretics	335 (86.6)	42 (89.4)	0.819	0.1
Beta-blockers	293 (75.7)	33 (70.2)	0.475	0.1
ACE-I/ARB	230 (59.4)	21 (44.7)	0.061	0.1
MRA	168 (43.4)	15 (31.9)	0.159	0.1
Laboratory data at discharge				
Haemoglobin (g/dL)	12.0 ± 2.0	11.5 ± 2.1	0.069	0.4
Creatinine (mg/dL)	1.19 (0.96–1.70)	1.18 (0.96–1.82)	0.940	0.0
BUN (mg/dL)	27 (20–36)	28 (18–41)	0.585	0.0
Serum sodium (mmol/L)	139 ± 4	138 ± 4	0.028	0.0
CRP (mg/dL)	0.27 (0.11–0.71)	0.18 (0.07–0.92)	0.192	2.3
BNP (pg/mL)	230 (123–469)	234 (108–493)	0.853	23.3

*Without respiratory sarcopenia' indicates patients who did not meet the definition of respiratory sarcopenia employed in this study; it does not imply the presence or absence of systemic sarcopenia.

Values are presented as the median (interquartile range), mean ± standard deviation, and *n* (%), as appropriate.

ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; BNP, brain natriuretic peptide; BUN, blood urea nitrogen; COPD, chronic obstructive pulmonary disease; CRP, C-reactive protein; DBP, diastolic blood pressure; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; NYHA, New York Heart Association; SBP, systolic blood pressure.

1.16; 95% CI, 0.66–2.01; *P* = 0.612), was associated with significantly higher 2 year mortality relative to the normal group (see [Supplementary material online, Table S4](#)).

Discussion

This *post hoc* analysis of the SONIC-HF registry provides novel insights into the prevalence and prognostic significance of respiratory sarcopenia in older patients hospitalized for heart failure. First, respiratory sarcopenia—defined by both reduced diaphragm thickness and low predicted FVC—was observed in ~10% of the cohort. Second, the presence of respiratory

sarcopenia was independently associated with an increased 2 year all-cause mortality, although neither of its individual components alone was associated with mortality, underscoring the potential clinical relevance of this composite condition for risk stratification.

Prevalence of respiratory sarcopenia

Several previous studies have attempted to determine the prevalence of respiratory sarcopenia. For instance, Chen *et al.* analysed 4301 older adults from the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative cohort and reported a 4 year cumulative incidence of 5.2% for

Table 2 Physical and pulmonary function measures stratified by respiratory sarcopenia

Variables	Without respiratory sarcopenia <i>n</i> = 388	Respiratory sarcopenia <i>n</i> = 47	<i>P</i> -value	Missing rate (%)
Physical function				
ASMI (kg/m ²)	7.0 [6.1–8.0]	6.7 [5.4–7.2]	0.021	43.9
Grip strength (kg)	20 [15–26]	16 [13–22]	0.001	11.1
Physical performance				
Gait speed (m/s)	0.8 [0.6–1.0]	0.7 [0.5–0.9]	0.007	7.5
Five-time chair-stand test (s)	12.0 [9.5–16.4]	12.8 [10.4–26.5]	0.047	16.3
SPPB	10 [8–12]	9 [5–11]	0.010	16.9
6MWD (m)	265 [169–343]	180 [100–271]	0.003	18.9
Pulmonary function				
VC (L)	2.3 [1.7–2.8]	1.4 [1.1–1.9]	<0.001	48.3
%VC (%)	83.2 [70.8–94.6]	65.9 [55.4–73.4]	<0.001	48.3
FVC (L)	2.1 [1.6–2.8]	1.4 [1.2–1.8]	<0.001	34.5
%FVC (%)	82.0 [68.7–94.1]	62.5 [52.8–71.0]	<0.001	34.5
FEV ₁ /FVC ratio (%)	73.0 [65.9–78.8]	74.1 [65.1–81.4]	0.584	34.5
%FEV ₁ (%)	83.6 [72.8–102.7]	72.8 [61.4–81.3]	<0.001	34.5

'Without respiratory sarcopenia' indicates patients who did not meet the definition of respiratory sarcopenia employed in this study; it does not imply the presence or absence of systemic sarcopenia.

Values are presented as the median (interquartile range).

6MWD, six-minute walk distance; ASMI, appendicular skeletal muscle mass index; FEV₁, forced expiratory volume in 1 s; %FEV₁, per cent predicted forced expiratory volume in 1 s; FVC, forced vital capacity; %FVC, per cent predicted forced vital capacity; SPPB, short physical performance battery; VC, vital capacity; %VC, per cent predicted vital capacity.

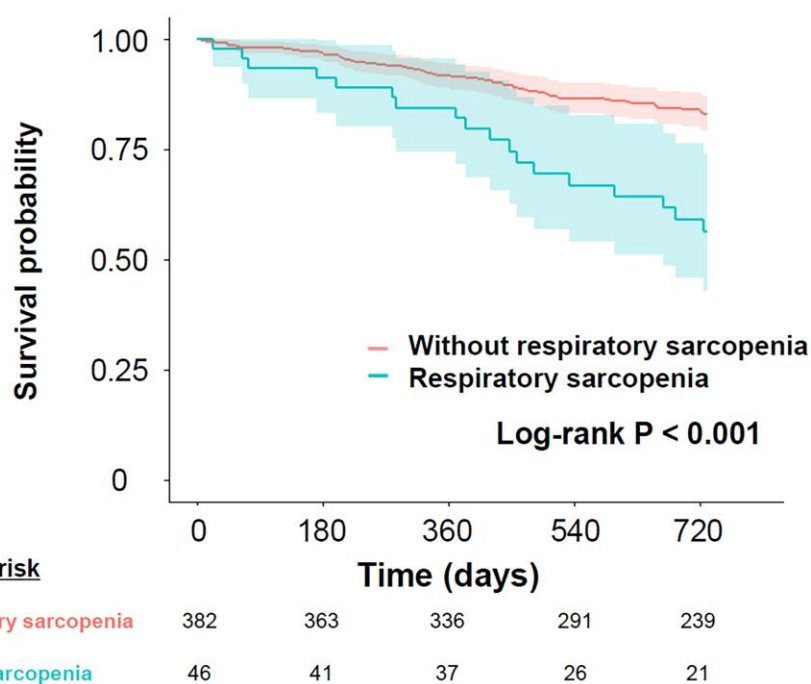


Figure 1 Kaplan–Meier curves for the primary outcome. Kaplan–Meier survival curves showing cumulative all-cause mortality over a 2 year follow-up period according to the presence or absence of respiratory sarcopenia. Shaded areas indicate 95% confidence intervals. Patients with respiratory sarcopenia exhibited significantly lower survival than those without (log-rank test, $P < 0.001$).

Table 3 Cox proportional hazards analysis of respiratory sarcopenia for 2 year mortality

	Unadjusted model			Adjusted model ^a		
	HR	95% CI	P-value	HR	95% CI	P-value
Without respiratory sarcopenia	1.00	(Reference)		1.00	(Reference)	
Respiratory sarcopenia	2.91	1.72–4.93	<0.001	2.51	1.40–4.50	0.002
MAGGIC risk score				1.08	1.04–1.13	<0.001
Log-transformed BNP				1.35	1.06–1.72	0.014

^aWithout respiratory sarcopenia' indicates patients who did not meet the definition of respiratory sarcopenia employed in this study; it does not imply the presence or absence of systemic sarcopenia.

^aAdjusted for the Meta-analysis Global Group in Chronic Heart Failure risk score and log-transformed brain natriuretic peptide.

BNP, brain natriuretic peptide; CI, confidence interval; HR, hazard ratio; MAGGIC, Meta-analysis Global Group in Chronic Heart Failure.

respiratory sarcopenia. In that study, respiratory muscle mass was assessed using appendicular skeletal muscle mass, and muscle strength was evaluated via peak expiratory flow rate (PEFR).²⁵ Similarly, Kera *et al.* conducted a 5 year follow-up of 470 community-dwelling older adults from the Otassha Study and reported different prevalence rates depending on the diagnostic definition—13.0% using PEFR-based criteria and 4.5% using the Japanese Association of Rehabilitation Nutrition (JARN) criteria, which define respiratory sarcopenia based on both respiratory muscle mass and maximal mouth pressure.^{12,26} Although definitions varied, the prevalence of respiratory sarcopenia in our heart failure cohort (10.8%) remained comparable to those in the general older population, supporting the validity of the findings. The slightly higher prevalence observed in this study compared with JARN-based diagnoses may partly be explained by differences in diagnostic criteria but may also reflect the impact of heart failure itself and the advanced age of the cohort, both of which contribute to respiratory muscle impairment.

Prognostic value of respiratory sarcopenia

A recent position paper established a conceptual definition of respiratory sarcopenia, identifying reductions in respiratory muscle mass and strength as its core components.⁵ Although specific cut-off values for these components remain undetermined, our study is among the first to demonstrate that these parameters can be used to diagnose respiratory sarcopenia and are prognostically relevant in older patients with heart failure. The prognostic value of respiratory sarcopenia has also been supported by previous research. For example, Sun *et al.* reported on 806 patients [61.7% men; median age: 69 (IQR: 64–76) years] who underwent lobectomy and mediastinal lymph node dissection for non-small cell lung cancer at two institutions in Japan between 2009 and 2018. In multivariable analysis, respiratory sarcopenia—defined based on PEFR and pectoralis muscle area—was independently associated with an increased risk of mortality, even after adjusting for several clinical variables, including age, sex, comorbidities, and cancer-related factors.²⁷ Although diaphragm thickness was not used in their diagnostic criteria, their findings reinforce our results that respiratory sarcopenia represents a significant prognostic factor.

Notably, this study further demonstrated that reduced diaphragm thickness alone, in the absence of reduced FVC, was not associated with increased mortality. Although decreased diaphragm thickness has been proposed as a risk factor for

adverse outcomes in ventilator-dependent patients—such as increased risk of reintubation and tracheostomy, prolonged duration of ventilatory support, and higher rates of respiratory complications—previous studies did not evaluate respiratory muscle strength, such as FVC.²⁸ Consequently, such studies may not have fully captured true respiratory sarcopenia, which is characterized by combined impairments in both respiratory muscle mass and strength. This finding aligns with observations in systemic sarcopenia, where functional measures such as gait speed and handgrip strength have been shown to be more predictive of clinical outcomes than muscle mass alone.^{29,30} Overall, our results emphasize the importance of incorporating both structural and functional assessments of respiratory muscles to define clinically and prognostically relevant respiratory sarcopenia.

A potential mechanism by which respiratory sarcopenia may affect prognosis is its linkage with multiple hormone deficiency syndrome (MHDS), which is frequently observed in patients with heart failure and has been associated with poor clinical outcomes.³¹ Circulating levels of anabolic hormones, including insulin-like growth factor-1, are often reduced in these patients, promoting skeletal muscle atrophy and contributing to the development of respiratory sarcopenia.^{32,33} Moreover, MHDS-related cardiopulmonary dysfunction may further aggravate respiratory muscle weakness, establishing a vicious cycle between hormonal deficiency and respiratory impairment.

Clinical implications

In this study, respiratory sarcopenia diagnosed via diaphragm thickness by ultrasonography and reduced FVC was significantly associated with all-cause mortality in older patients with heart failure.

Physical frailty is most commonly operationalized using the Fried phenotype, which defines frailty based on five criteria: unintentional weight loss, self-reported exhaustion, weakness, slowness, and low physical activity.^{34,35} Respiratory sarcopenia, characterized by reduced respiratory muscle strength and function, may contribute to key components of this phenotype, particularly self-reported exhaustion and reduced physical activity, through impaired exercise tolerance. Although systemic sarcopenia of the appendicular musculature has traditionally been emphasized as a major contributor to physical frailty, our findings suggest that respiratory sarcopenia may also be relevant to the development of frailty as defined by the Fried

phenotype.³⁶ Therefore, when physical frailty is identified using the Fried criteria, respiratory sarcopenia should be considered among the potential underlying conditions in older patients with heart failure.

Unlike cardiac ultrasonography, measuring diaphragm thickness is relatively straightforward because the diaphragm exhibits less pulsatile movement and provides a stable imaging window. As demonstrated in the SONIC-HF study, even non-cardiac specialists, such as physical therapists, were able to reliably acquire diaphragm images after brief training.⁹ In contrast to whole-body sarcopenia assessments that require BIA, which is contraindicated in patients with pacemakers, or dual energy X-ray absorptiometry, diaphragm thickness can be measured safely and non-invasively in this population. Although techniques such as maximal inspiratory pressure and sniff nasal inspiratory pressure can assess respiratory muscle strength, they require specialized equipment and standardized procedures and are influenced by device and mouthpiece design.³⁷ FVC, by comparison, is a routine pulmonary function test that can be quickly administered by trained technicians using widely available equipment. Moreover, although conventional sarcopenia diagnosis requires multiple physical performance tests, such as SPPB, gait speed, or the five-time sit-to-stand test, respiratory sarcopenia can be identified using just two simple and feasible parameters.⁶ Importantly, early bedside assessment based on respiratory sarcopenia enables clinicians to identify high-risk patients at the earliest stages of hospitalization, thereby creating a critical window for initiating interventions before irreversible decline occurs. Evidence from prior studies suggests that combining supervised in-hospital ambulation with early cardiac rehabilitation can prevent deconditioning, enhance functional recovery, and, in some cases, exceed baseline physical capacity.^{38–40} This aligns with findings from the ACTIVE-ADHF randomized controlled trial, which demonstrated that in-hospital, exercise-based cardiac rehabilitation significantly improved 6MWD, physical and cognitive function, and quality of life at discharge, without compromising safety.⁴¹ Combined, these observations suggest that recognizing suitable patients early and initiating timely, targeted interventions could facilitate tailored cardiac rehabilitation with substantial short-term benefits that may translate into improved long-term outcomes.

Limitations

First, as a *post hoc* analysis of data derived from a prospective observational cohort, the methodology was inherently exploratory. Therefore, the findings should be interpreted as hypothesis-generating rather than definitive. Although adjustments were made for established covariates, the influence of unmeasured confounders, such as the severity of chronic lung disease, nutritional status, systemic frailty indices, and underlying musculoskeletal or respiratory conditions, could not be fully excluded. These factors may have affected both the development of respiratory sarcopenia and subsequent clinical outcomes. Second, the diagnostic thresholds for respiratory sarcopenia were based on a recently proposed position paper and prior literature. However, these cut-off values have not been fully validated in patients with heart failure, which may have affected the sensitivity and specificity of the classification. Third, respiratory sarcopenia was assessed only once at the time

of hospital discharge. Serial or longitudinal assessments during recovery or disease progression might provide more comprehensive prognostic insights but were not performed in this study. Fourth, hormonal profiles related to sarcopenia and heart failure were not assessed in this study. Fifth, only resting measurements of pulmonary function were performed; consequently, spirometry data were unavailable, precluding assessment of respiratory function during exercise. Sixth, some patients were excluded owing to missing diaphragm or respiratory function data (e.g. FVC), which may have biased prevalence estimates and risk associations towards healthier or more testable patients. Moreover, although the presence of COPD and additional spirometric parameters (FEV₁/FVC ratio and % FEV₁) was adjusted for in the sensitivity analysis, other pulmonary factors, such as the diffusing capacity of the lung for carbon monoxide, COPD exacerbations, and COPD-related pharmacotherapy (e.g. inhalation therapy or corticosteroid use), were not available in the SONIC-HF registry. Therefore, residual confounding by unmeasured pulmonary variables cannot be completely excluded. Seventh, sodium-glucose cotransporter 2 (SGLT2) inhibitors had not yet been established as standard therapy for diabetes or heart failure during the registry enrolment period; therefore, we could not assess whether SGLT2 inhibitors modify the clinical impact of respiratory sarcopenia. Finally, because the study cohort consisted exclusively of older Japanese patients with heart failure, generalizability to younger, non-Asian, or more diverse populations may be limited owing to potential differences in body composition, sarcopenia prevalence, and respiratory function norms.

Conclusions

This *post hoc*, retrospective analysis of a multicentre cohort of older patients with heart failure demonstrated that respiratory sarcopenia, defined by concurrent low diaphragm thickness and reduced FVC, was relatively uncommon but independently associated with increased 2 year all-cause mortality. These findings highlight the prognostic relevance of assessing both respiratory muscle mass and function rather than structural measures alone. Incorporating respiratory sarcopenia screening into routine clinical assessments may enhance risk stratification and guide targeted management strategies in this vulnerable population.

Supplementary material

Supplementary material is available at [European Journal of Preventive Cardiology](#).

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Author contributions

Taisuke Nakade (Conceptualization [lead]; Data curation [equal]; Formal analysis [equal]; Methodology [equal]; Project administration [equal]; Writing—original draft [lead]; Writing—review & editing [lead]), Daichi Maeda (Conceptualization [lead]; Formal analysis [lead]; Supervision [lead]; Writing—original draft [lead];

Writing—review & editing [lead]), Yuya Matsue (Conceptualization [lead]; Data curation [lead]; Formal analysis [lead]; Funding acquisition [lead]; Investigation [lead]; Methodology [lead]; Project administration [lead]; Resources [lead]; Software [lead]; Supervision [lead]; Validation [supporting]; Visualization [lead]; Writing—original draft [supporting]; Writing—review & editing [lead]), Nobuyuki Kagiya (Conceptualization [lead]; Data curation [lead]; Methodology [lead]; Supervision [supporting]; Writing—review & editing [supporting]), Yoshiaki Ikeda (Data curation [supporting]; Writing—review & editing [supporting]), Kazuya Saito (Conceptualization [supporting]; Formal analysis [supporting]; Investigation [supporting]; Methodology [supporting]; Visualization [supporting]; Writing—review & editing [supporting]), Kentaro Kamiya (Conceptualization [lead]; Data curation [lead]; Funding acquisition [equal]; Supervision [equal]; Writing—review & editing [supporting]), Hiroshi Saito (Data curation [equal]; Writing—review & editing [supporting]), Misako Toki (Data curation [supporting]; Writing—review & editing [supporting]), Emi Maekawa (Data curation [supporting]; Writing—review & editing [supporting]), Kenji Yoshioka (Conceptualization [supporting]; Data curation [supporting]; Writing—review & editing [supporting]), Takeshi Kitai (Conceptualization [lead]; Data curation [supporting]; Writing—review & editing [supporting]), Kentaro Iwata (Data curation [supporting]; Writing—review & editing [supporting]), Azusa Murata (Data curation [supporting]; Writing—review & editing [supporting]), Akihiro Hayashida (Data curation [supporting]; Writing—review & editing [supporting]), and Tohru Minamino [Writing—review & editing (supporting)]

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Data availability

The data supporting this study's findings are available from the corresponding author, Yuya Matsue, upon reasonable request.

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